AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

Physics Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL NINE questions. Questions 7, 8 & 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Complete the given table of quantities.

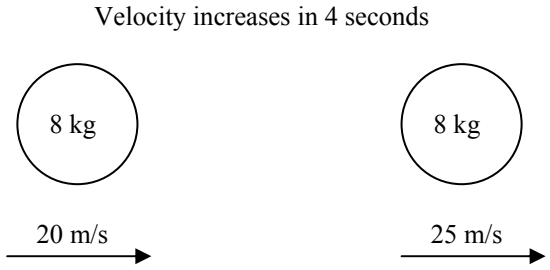
Fundamental and derived units

Quantity	Formula	SI unit	Measuring Instrument
		Ampere	
			Spring balance
Heat			

Q.2. (Total 6 Marks)

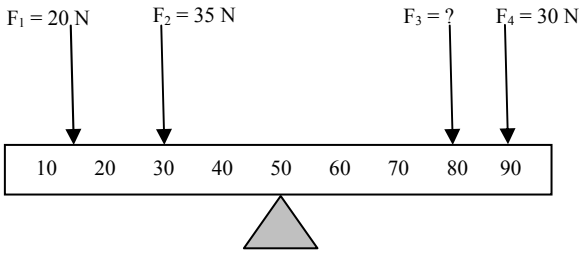
a. What are the TWO different ways of deriving the formula of Newton's second law of motion? Prove their equivalence. (3 Marks)

b. Calculate the force that must be applied in the given diagram. (3 Marks)



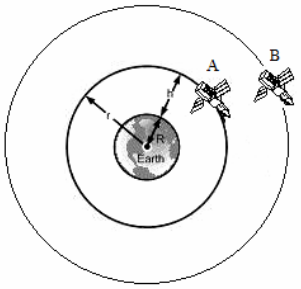
Q.3. (Total 3 Marks)

State the principle of moments and find the magnitude of F_3 in the given diagram.



Q.4. (Total 3 Marks)

The given diagram shows two identical satellites *A* and *B* orbiting the earth.



a. Which one of the satellites is attracted more strongly by the earth? (1 Mark)

_____

_____

b. Which of the satellites is revolving with a slower speed and why? (2 Marks)

_____

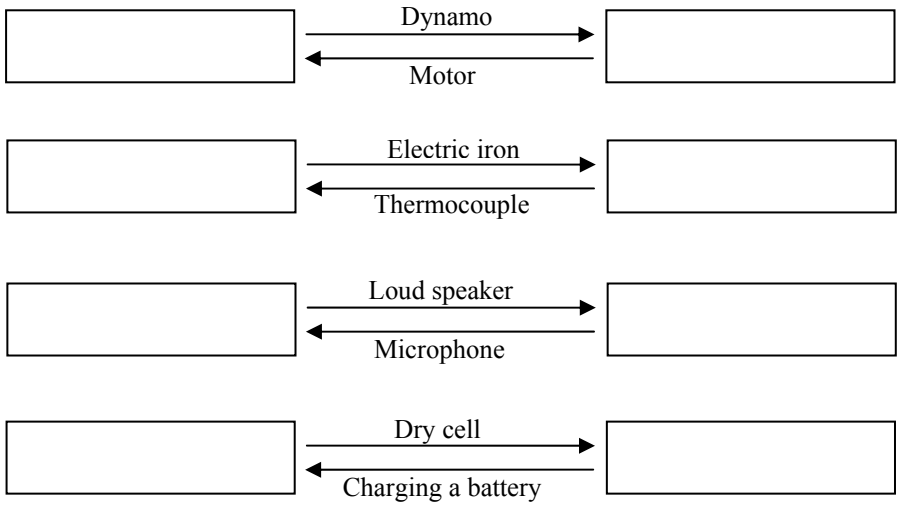
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Q.5. (Total 6 Marks)

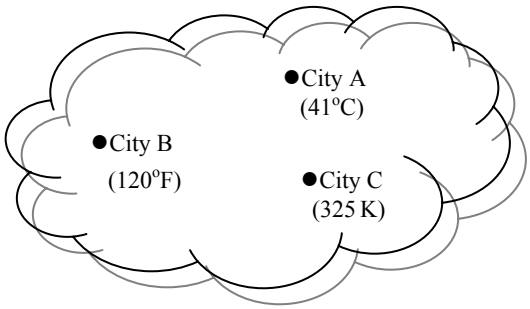
a. In the given diagram, one form of energy is converted into the other form using given hints between the blocks. Fill the blocks with correct form of energy. (4 Marks)



b. Define any TWO forms of energy. (2 Marks)

Q.6. (Total 4 Marks)

Temperatures of three different cities in summer are given. Which one of the cities will be facing more heat? Show the calculation.



City A	City B	City C

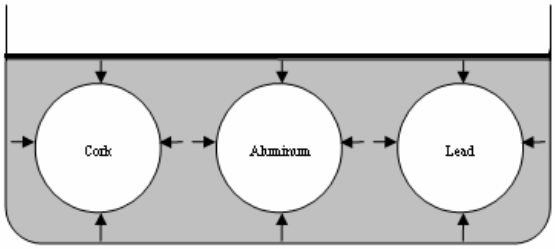
Q.7. (Total 5 Marks)

EITHER

a. Explain the effect of atmospheric pressure with the help of FIVE examples from daily life.

OR

b. The given diagram represents three balls of equal volume with specific gravities 0.2, 2.7 and 11.3 respectively which are immersed in water.



Explain what is meant by buoyant force and what is the effect of it on each ball? Describe the behaviour of each ball after the effect of buoyant force.

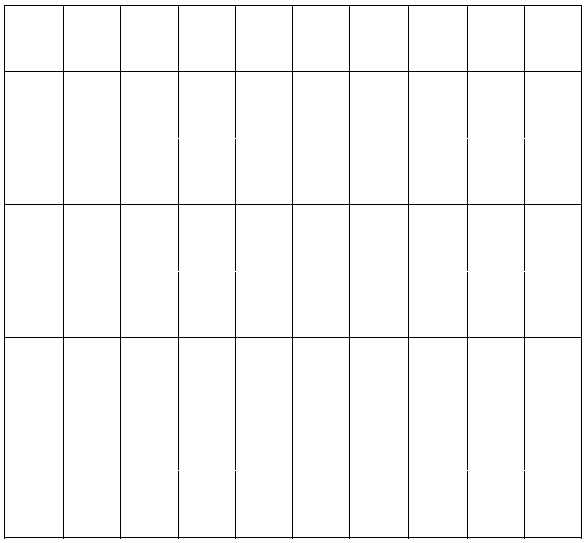
Q.8. (Total 5 Marks)

EITHER

a. A locomotive is pulling a train from one station and travels along a straight horizontal track towards another station. Its speed at regular interval of time is recorded in the given table.

Velocity	0	30	40	40	40	40	0
Time	0	1	2	4	6	8	12

Draw and describe the graph of the whole journey in terms of uniform velocity, acceleration, and retardation.



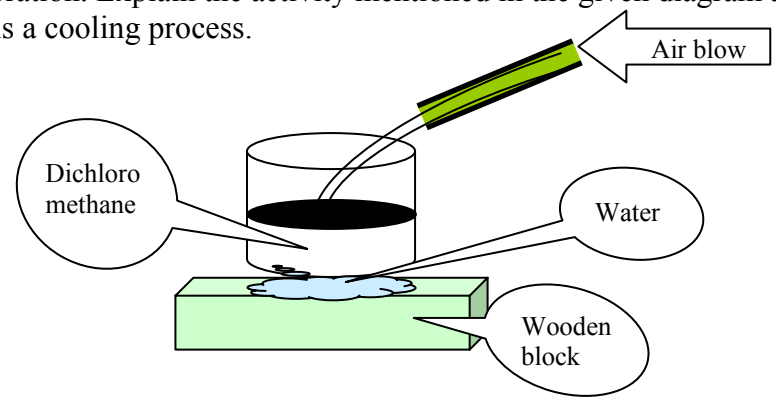
OR

b. If a bullet of mass 10 gm is fired into a block of wood of mass 390 gm lying on a smooth surface, the block moved at a velocity of 10 m/s. What will be the velocity of the bullet?

Q.9. (Total 5 Marks)

EITHER

a. Define evaporation. Explain the activity mentioned in the given diagram and prove that evaporation is a cooling process.



OR

b. State any FIVE examples of heat transfer by the process of convection in everyday life.

END OF PAPER

Please use this page for rough work

Please use this page for rough work